

Index

- absence seizures, 183
- absence status epilepticus, 234
- absences with eyelid myoclonia, 183
- activation procedures, 65, 160–168
- active reference, 27, 29–31
- active sleep, neonates/preterm births
 - and, 46
- active state, 42
- activite moyenne, 45, 46
- age-dependent epilepsy syndromes, 187
- alpha activity, 32
- alpha coma, 224
- alpha rhythm, 33–36
 - abnormalities of, 35
 - normal variations of, 33
- alpha squeak, 33
- alpha-theta coma, 224
- alpha variants, 35
- amplifiers, 7
- amplitude, 32
 - of the alpha rhythm, 34
 - of the background cerebral activity, 71
 - symmetry and, 67
 - of waveforms, 80
- Angelman syndrome, 199
- anterior dysrhythmia, 45
- anterior posterior gradient (APG), 73
- anti-NMDA receptor encephalitis, 226
- antiseizure medications, 187
- apnea in infants, polygraphs and, 41
- arrythmias
 - in infants, polygraphs and, 41
 - systemic seizure mimics and, 174
- artifacts, 64, 83–97
 - external artifacts and, 83
 - internal artifacts and, 86–96
- artificial potentials, 20, 27, 29–31
- asymmetric alpha rhythm, 35
- asymmetry, 67
- asynchrony, 43
- atonic seizures, 180
- atonic status epilepticus, 236
- attenuation, 44, 69
 - drowsiness and, 53
 - electrographic seizures and, 152
- atypical absence status epilepticus, 235
- atypical absences, 183
- autosomal dominant nocturnal frontal lobe epilepsy (ADNFLE), 206
- autosomal dominant temporal lobe epilepsy (ADTLE), 207
- average reference montage, 17
- awareness
 - focal seizures and, 178
 - generalized seizures and, 179
- Bancaud's phenomena, 35, 214
- barbiturates, excessive beta and, 37
- benign epilepsy with centrotemporal spikes (BECTS), 27, 199
- benign epileptiform transients of sleep (BETS), 100
- benign infantile seizures, 197
- benign neonatal convulsions, 196
- benign post cardiac arrest patterns, 242
- benign sporadic sleep spikes (BSSS), 100
- benign temporal delta transients of the elderly, 115
- benzodiazepine overdose, 225
- benzodiazepines
 - de novo absence status epilepticus and, 235
 - excessive beta and, 37
- beta, asymmetric beta and, 37
- beta activity, 32
- beta coma, 225

- beta rhythm, 36
- bilateral independent (BI) pattern, 136
- bilateral independent periodic discharges (BIPDs), 144
- bilateral tonic clonic seizures, 179
- bio calibration, 11
- bipolar montages, 17
 - localization and, 23, 24
 - strengths/weaknesses of, 19
- brain death, 246–247
- breach effect, 37, 61, 216
- brief potentially ictal rhythmic discharges (BIRDs), 157, 184
- calibration signal, 11, 61
- cardiac arrest. *see* post cardiac arrest
- cardiac artifacts, 91
- cardioballistogram artifacts, 93
- central reference (Cz), 29
- cerebral activity, 64
- cerebral dysfunction, delta rhythm and, 38
- cerebral lesions, delta rhythm and, 38
- channels, 3, 9, 15, 17
- chewing artifacts, 94
- childhood absence epilepsy (CAE), 202
 - generalized epilepsies and, 192
 - generalized spike and wave discharges and, 124
 - hyperventilation and, 161
 - occipital intermittent rhythmic delta activity and, 138
- children, 45–51
 - absence status epilepticus and, 235
 - atonic status epilepticus and, 236
 - clonic status epilepticus and, 236
 - EEG maturation and, 45–51
 - epilepsy syndromes and, 195, 199–206
 - myoclonic status epilepticus and, 236
 - physiological states/sleep and, 49
 - recording EEGs in, 40
- chronological age, 42
- Ciganek rhythm, 110
- circumferential montage, 17
- clonic status epilepticus, 236
- collodion, 11
- conceptual age (CA), 42
- continuity, 43, 69
- continuous spike and wave during sleep (CSWS), epileptic encephalopathy and, 204
- contralateral ear reference, 31
- contralateral ear reference montage, 17
- convulsive status epilepticus (CSE), 153, 230, 231
- coronal montage, 17
- cortical myoclonus, 236
- craniotomy, as a consideration when reading an EEG, 61
- Creutzfeldt-Jakob disease (CJD), 12, 225
- ctenoids, 101
- dacrycitic seizures, 209
- de novo absence status epilepticus, 235
- delta activity, 32
- delta brushes, 45
- delta rhythm, 38
- delta slowing, 128
- DEPDC5 gene mutation, 209
- diffuse cortical dysfunction, low background voltage and, 72
- diffuse delta, 38
- digital displays, 8, 9, 18, 31
- discontinuity, 43, 70
- disposable electrodes, 12
- Doose syndrome, 203
- Dravet syndrome, 192, 203
- drowsiness
 - adults and, 53
 - infants and, 48
- duration, of waveforms, 80
- dysormia, 56
- dysmaturity, 42, 43
- EEG (electroencephalogram), 1–14
 - an approach to reading, 61–66
 - components of, 4, 7
 - cortex and, 2
 - historical context of, 2
 - indications for, 3

- interpreting in neonates, 42
- limitations of, 3
- potentials and, 2
- writing reports and, 248–252
- EKG artifacts, 91
- EKG channel, 65
- electrical status epilepticus in sleep (ESES), 204
- electro cerebral inactivity (ECI), 44
- electrocardiogram artifacts, 91
- electrocerebral inactivity (ECI), 246
- electrode artifacts, 83
- electrode potentials, 15
- electrodes, 4
 - disposable, 12
 - International 10–20 system of electrode placement and, 5
 - placing, 4–7, 8, 40
- electrographic activity, 2, 32, 43
- electrographic seizures, 152
 - criteria for determining, 153
 - rhythmic variants and, 105
- encephalitis, multifocal epilepsies and, 192
- encephalopathy, 219–229
- enconches fontanelles, 45
- end of chain, in bipolar montage, 27
- epilepsia partialis continua (EPC), 234
- epilepsy, 187–194, 198, 202, 207
 - classification of, 177, 187
 - diagnosing, 187
 - epileptiform abnormalities and, 4
 - epileptogenic abnormalities and, 65
 - frontal discharges and, 122
 - migraines and, 170
 - normal variants and, 98
 - types of, 188–193
- epilepsy of infancy with migrating focal seizures, 198
- epilepsy syndromes, 195
 - in adolescents and adults, 196, 206–208
 - in children, 195, 199–206
 - in infants, 195, 197
 - in neonates, 195, 196
 - less age-specific, 196, 209
- epilepsy with generalized tonic clonic seizures alone (EGTCS), 207
- epilepsy with myoclonic absences (EMA), 202
- epileptic discharges, key features of, 120
- epileptic encephalopathy with continuous spike and wave during sleep (CSWS), 204
- epileptic seizures, 65, *see also* seizures
 - electrographic seizures and, 152
 - epileptogenic abnormalities and, 65
 - lateralized periodic discharges and, 143
 - non-epileptic seizures and, 174
 - plus terms and, 150
 - seizure mimics and, 169–176
- epileptic spasms, 181
- epileptiform abnormalities, 4
- epileptogenic abnormalities (discharges), 65
- epileptogenicity, 216
- evolution, 65
 - electrographic seizures and, 152
 - ictal patterns and, 149
- extreme delta brush, 226
- eye artifacts, 86
- eyelid myoclonia with absences (EMA), 183
- familial focal epilepsy with variable foci (FFEVF), 209
- familial mesial temporal lobe epilepsies (FMTLE), 207
- fast ripples, 32
- fifth day fits, 196
- filters, 10, 61
- fluctuating patterns, 149
- focal delta, 38
- focal discharges, 24, 122, 124
- focal dysfunction
 - detecting, 214
 - estimating the severity of, 215
 - physiologic focal dysfunction and, 215

- focal epilepsy, 188
 - in children, benign epilepsy with centrottemporal spikes and, 199
 - focal discharges and, 122
 - lateralized periodic discharges and, 143
 - subclassifications of, 188
- focal motor seizures, lateralized periodic discharges and, 143
- focal non-motor seizures, 189
- focal potentials, bipolar montages and, 19, 23
- focal seizures, 178, 192
- focal to bilateral convulsive status epilepticus, 231
- frequency, 32
 - organization and, 73
 - symmetry and, 67
 - of waveforms, 81
- frontal arousal rhythm (FAR), 105
- frontal lobe epilepsy, 188
- gain, 8
- Gastaut syndrome, 201
- gelastic seizures, associated with
 - hypothalamic hamartoma, 196, 209
- generalized convulsive status epilepticus (GCSE), 231
- generalized discharges, 124
- generalized epilepsy, 192, 234
- generalized epilepsy with febrile seizures plus (GEFS+), 203
- generalized (G) pattern, 135
- generalized periodic discharges (GPDs), 79, 136, 138–143
 - Creutzfeldt-Jakob disease and, 225
 - encephalopathies and, 221
 - subacute sclerosing panencephalitis and, 226
- generalized potentials, 20, 23
- generalized rhythmic delta activity (GRDA), 137
 - cardioballistogram artifacts and, 93
 - eye artifacts and, 88
 - tongue artifacts and, 91
- generalized seizures, 179–184, 192
- generalized spike and wave discharges (GSW), 124, 135
- generalized tonic clonic seizure (GTCs), 180
- generalized tonic status epilepticus, 235
- genetic generalized epilepsy (GGE), 192
- gestational age, 42
- global dysfunction, 219–229
- glossopharyngeal artifacts, 91
- glucose transporter (GLUT1) deficiency, 202
- graphoelements, 42, 45
- GRIN2A mutation, 204
- head tremor artifacts, 94
- headbox, 8
- hemiconvulsion-hemiplegia epilepsy (HHE), 196, 209
- hemiplegia, 209
- high-frequency filter (HFF), 10
- high-frequency oscillations (HFOs), 32
- high-pass filter (HPF), 10
- highly epileptiform bursts (HEBs), 71
- highly malignant EEG, 240
- hippocampal sclerosis, 207
- horizontal dipole, 27
- hypersynchrony, 48, 49, 53
- hyperventilation (Hv), 65, 160
 - abnormal responses and, 161
 - contraindications for, 160
 - intermittent photic stimulation and, 163
- hypnopompic hypersynchrony, 48
- hypoglycemia, hyperventilation and, 162, 174
- hypothalamic hamartoma (HH), gelastic seizures and, 196, 209
- hypotonia, 56
- ictal activity, 65, 71
- ictal patterns, 149–159

- idiopathic generalized epilepsy (IGE), 192, 198
- ILAE classification of epilepsies, 177
 - epilepsy diagnosis and, 187
 - status epilepticus and, 230
- impedances, 8
- indeterminate sleep, neonates/preterm births and, 47
- infantile spasms, 198
- infants
 - epilepsy syndromes and, 195, 197
 - physiological states/sleep and, 47
- infra-slow oscillations (IFOs), 32
- insular epilepsy, 191
- interictal epileptic discharges, 99, 105
- International 10–20 system of electrode placement, 5
- intracerebral hemorrhage (ICH), lateralized periodic discharges and, 143
- ipsilateral ear reference, 31
- ipsilateral ear reference montage, 17
- isopotentials, 24
- juvenile absence epilepsy (JAE), 192, 207
- juvenile myoclonic epilepsy (JME), 192, 207
- K complexes, 55, 75
- Lafora body disease, 209
- lambda waves, 113
- Landau-Kleffner syndrome (LKS), 204
- late childhood onset occipital epilepsy, 201
- lateral temporal lobe seizures (LTLS), 189
- lateralized (L) pattern, 135
- lateralized periodic discharges (LPDs), 143
 - encephalopathies and, 221
 - plus terms and, 150
- lateralized rhythmic delta activity (LRDA), 138, 216
- Lennox-Gastaut syndrome (LGS), 192, 205, 235
- lesions, 214–218
- levetiracetam, 142
- LGI1 defect, 207
- localization, 23–28
- locked-in syndromes, 225
- log notes, 65
- longitudinal bipolar (double banana) montage, 9, 17
- low-frequency filter (LFF), 10
- low-pass filter (LPF), 10
- malignant EEG, 240
- maturation, 40–52
 - basics of interpretation and, 42
 - principles of, 43
- mesial temporal lobe epilepsy (MTLE), 188
- mesial temporal sclerosis (MTS), 177
- metabolic disturbances, systemic seizure mimics and, 174
- metabolic encephalopathy, triphasic waves and, 140–143
- midline theta, 38, 110
- migraines, 170
- migrating partial seizures of infancy, 198
- mitten waves, 56
- mixed epilepsy, 192
- monorhythmic delta, 45
- montages, 3, 9, 17–22
 - checking, 61
 - digital displays and, 31
 - localization and, 23–28
- motor seizures
 - focal, lateralized periodic discharges and, 143
 - frontal lobe epilepsy and, 188
 - seizure onset and, 179
- movement disorders, 174
- moyamoya disease, hyperventilation and, 162
- mu rhythm, 37, 110
- multifocal epilepsies, 192
- multifocal (Mf) pattern, 136

- muscle artifacts, 53–56, 93
- myoclonic absences, 183
- myoclonic atonic/astatic epilepsy (MAE), 203
- myoclonic-atonic seizures, 180
- myoclonic encephalopathy in
 - non-progressive disorders, 199
- myoclonic epilepsy
 - infants and, 198
 - progressive myoclonic epilepsies and, 208
- myoclonic seizures, 180
- myoclonic status epilepticus, 236, 243
- myoclonus, myogenic artifacts and, 94
- myogenic artifacts, 93, 179–182

- neocortical epilepsy, 189
- neonates
 - epilepsy syndromes and, 195, 196
 - interpreting EEGs and, 42
 - physiological states/sleep and, 45–47
 - polygraphs and, 41
 - preterm, infra-slow oscillations and, 32
- neurological seizure mimics, 170–174
- neuronal ceroid lipofuscinosis (NCL), 166, 209
- neuronal injury
 - non-convulsive seizures and, 157
 - periodic discharges and, 135
- NMDA encephalitis, 226
- nocturnal seizures, 173
 - epilepsy diagnosis and, 187
 - frontal lobe epilepsy and, 188
 - sleep disorders and, 173
- non-convulsive seizures (NCS), 149, 152
- non-convulsive status epilepticus (NCSE), 149, 152, 230, 233
 - diagnosing, 223
 - generalized periodic discharges and, 139
- non-epileptic seizures, 174
- non-motor onset seizures, focal, 189
- non-rapid eye movement sleep (NREM), 54
 - normal variants, 64, 98–118
 - 14 and 6 Hz Positive spikes and, 101
 - 6 Hz spike and wave discharges and, 103
 - types/features of, 98
 - notch filter, 10
- occipital intermittent rhythmic delta activity (OIRDA), 65, 138
- occipital lobe epilepsy, 191
- occipital references (Oz), 31
- occipital spikes, 123
- Ohtahara syndrome, 196
- organization, of the background cerebral activity, 73

- Panayiotopoulos syndrome, 200
- panic attacks, 175
- paper speed, 10
 - checking, 61
 - EEG recording in early life and, 41
- paradoxical alpha rhythm, 35
- parietal lobe epilepsy, 189
- periodic discharges, 136
- phantom spike and wave, 103
- phase reversals, 23–28
- photic stimulation (Ps), 65, 163–167
 - abnormal responses and, 163
 - artifactual responses and, 166
 - contraindications for, 163
- photoparoxysmal response (PPR), 163
- photosensitive epilepsies, 163
- physiologic focal dysfunction, 215
- plus features/plus terms, 150
- polarity, 15
- polygraphs, neonatal EEGs and, 41, 42
- positive bursts of 14 Hz and 6 Hz, 101
- positive occipital sharp transients of sleep (POSTS), 55, 113
- post anoxic coma, 71
- post cardiac arrest, 239–245
 - electrographic patterns and, 240
 - seizures/status epilepticus and, 242
- post menstrual age (PMA), 42

- posterior dominant rhythm (PDR), 33
 in adults, 53
 background cerebral activity and, 73
 in children, 49
- posterior slow waves of youth (PSWY), 115
- posterior variants, 113–116
- preterm neonates
 infra-slow oscillations and, 32
 physiological states/sleep and, 45–47
- prevalence, of waveforms, 79
- prion disease, 12
- psychological seizure mimics, 174
- psychomotor variant, 106
- quiet sleep, neonates/preterm births
 and, 46
- quiet state, 42
- rapid eye movement (REM), 54, 56, 173
- Rasmussen's syndrome, 196, 209
- reactivity, 44, 73
- reference electrode, referential montages
 and, 29
- referential montages, 17
 localization and, 23, 27
 mitten waves and, 56
 strengths/weaknesses of, 21
- reflex epilepsies, 209
- REM sleep, 54, 56, 173
- repetitive abnormalities, 134–148
 an approach for classifying repetitive
 patterns of, 135
 types of, 134
- reports, writing, 248–252
 main parts to include, 248
 sample interpretations and, 251
 sample reports and, 249–251
- reticular activating system (RAS), 2
- rhythmic artifacts, 86
- rhythmic delta activity (RDA), 136
- rhythmic mid-temporal theta of
 drowsiness (RMTD), 106
- rhythmic variants, 105–113
- rhythms, 32–38
- ripples, 32
- Rolandic epilepsy, 199
- safety and safety standards, 11
- sampling frequency, 8
- sawtooth waves, 56
- scalp electrodes. *see* electrodes
- scalp potentials, 2
- SCN1A defect, 203
- sedatives, excessive beta and, 37
- seizure mimics, 169–176
 common pediatric behavioral
 conditions and, 175
 types of, 169
- seizures, 65, 177–186, *see also* epileptic
 seizures
 classification of, 177, 187
 epilepsy diagnosis and, 187
 key observations of semiology and, 178
 in neonates, 184
 non-epileptic, 174
 post cardiac arrest and, 242
- sensitivity, 10
 checking, 61
 EEG recording in early life and, 40
- severe myoclonic epilepsy, 203
- sharp transients, 121
- sharp variants, 99
- sharp waves, 120–125
- SI (stimulus induced) prefix, 136
- skin erosion, 11
- skull defect, 37, 61, 216
- sleep, 43, 54–58
 EEG recording in early life and, 41
 in infants, 48
 neonates/preterm births and, 45
 transitional, 45
- sleep disorders, 173
- sleep spindles, 55, 75
- sleep-wake cycle, 41–45
 infants and, 49
 maturation of, 44
- sleeping aids, de novo absence SE
 and, 235
- slow wave sleep, 43, 45, 56

- slow waves, 127, 128
- small sharp spikes (SSS), 101
- spike and wave (SW) pattern, 136
- spindle coma, 224
- sporadic abnormalities, 119–133
 - repetitive abnormalities and, 134
 - types of, 119
- square wave calibration, 11
- stage I sleep, 55
- stage II sleep, 55, 75
- stage III sleep, 56
- static patterns, 149
- status epilepticus (SE), 71, 230–238
 - post cardiac arrest and, 242
 - types of, 230, 234
- stimulus induced (SI) prefix, 136
- structural dysfunction, 214, 215
- subacute sclerosing panencephalitis (SSPE), 226
- subarachnoid hemorrhage (SAH),
 - lateralized periodic discharges and, 143
- subclinical rhythmic electrographic (theta) discharges of adults (SREDA), 108
- subcortical myoclonus, 236
- subcortical structures, 2
- subdivisions of EEG signals, 9
- subdural hematoma, asymmetric beta and, 37
- suppression, 44, 69, 152
- sweat artifacts, 84
- symmetry, 67
- synchrony, 43
- syncopes, 170
- technologist's log notes, 65
- temporal discharges, 31
- temporal intermittent rhythmic
 - delta activity (TIRDA), 65, 128, 138, 216
- temporal lobe epilepsy, 188
- temporal lobectomy, as a consideration
 - when reading an EEG, 61
- temporal predominant lateralized
 - rhythmic delta activity, 128
- texting rhythm, 111
- thalamus, 2
- theta activity, 32
- theta rhythm, 37, 110
- time base, 61
- tongue artifacts, 91
- tonic seizures, 180
- tonic status epilepticus, 235
- toxic disturbances, systemic seizure
 - mimics and, 174
- toxic encephalopathy, triphasic waves
 - and, 140–143
- trace alternans, quiet sleep and, 43, 46, 48
- trace discontinu, 43
- transient ischemic attacks (TIA), 170
- transitional sleep, 45, 47
- transverse bipolar montage, 17
- tremors (head), myogenic artifacts
 - and, 94
- triphasic waves, 140–143
 - anti-epileptic medications trial and, 142
 - electrographic features of, 141
 - key identifying features of, 140
- typical absence status epilepticus, 234
- typical absences, 183
- Unverricht-Lundborg syndrome (UVL), 209
- variability, 44, 73
- ventilator artifacts, 85
- vertex reference montage, 17
- vertex waves, 55
- voltage, 44
 - of the alpha rhythm, 34
 - of the background cerebral activity, 71
- wakefulness state, 42, 53
 - infants and, 48
 - neonates/preterm births and, 45
- West syndrome, 198
- wicket waves, 100